

Topic: Memory Banking in 8086

Q. What is Memory Banking in the 8086 microprocessor? Explain the roles of the A0 and BHE' signals in accessing Even and Odd memory banks for byte and word transfers.

 Definition to Remember

Memory Banking is the hardware organization of the 8086's 1 MB physical memory into two parallel 512 KB halves: the **Even Bank** (connected to D0-D7) and the **Odd Bank** (connected to D8-D15). This allows the CPU to efficiently read/write either 8-bit bytes or 16-bit words.

1. Concept of Memory Banking

To support a 16-bit data bus while maintaining byte-level addressing, memory is split:

- * **Even Bank (Lower Bank):** Contains even addresses (00000H , 00002H). Connected to the lower half of the data bus (**D0 - D7**).
- * **Odd Bank (Upper Bank):** Contains odd addresses (00001H , 00003H). Connected to the upper half of the data bus (**D8 - D15**).

2. Role of A0 and BHE' Signals

The CPU uses two specific signals to select which bank to activate during a memory cycle:

- * **A0 (Address Line 0):** Selects the **Even Bank** (active when A0 = 0).
- * **BHE' (Bus High Enable):** Selects the **Odd Bank** (active low, BHE' = 0).

3. Byte and Word Access Scenarios

BHE'	A0	Selection	Transfer Type
0	0	Both Banks	Read/Write 16-bit Word from an Even Address. Uses D0-D15. Completes in 1 Clock Cycle .
0	1	Odd Bank	Read/Write 8-bit Byte from an Odd Address. Uses D8-D15.
1	0	Even Bank	Read/Write 8-bit Byte from an Even Address. Uses D0-D7.
1	1	None	No memory access.

4. Accessing a Word from an Odd Address (Unaligned Access)

If the CPU tries to read a 16-bit word starting at an **odd address** (e.g., 00001H), it cannot do it in one cycle. It requires **two clock cycles**:

- * **Cycle 1 (BHE'=0, A0=1):** Reads the lower byte from the odd address (00001H) via D8-D15.
- * **Cycle 2 (BHE'=1, A0=0):** Increments address to the next even boundary (00002H) and reads the upper byte via D0-D7.

★ Must-Write Points (for 10 marks)

1. **Memory Banking** splits 1 MB memory into two 512 KB banks: **Even** and **Odd**.
2. **Even Bank**: Even addresses, connected to lower data bus (**D0-D7**).
3. **Odd Bank**: Odd addresses, connected to upper data bus (**D8-D15**).
4. **A0=0** enables the Even Bank; **BHE'=0** enables the Odd Bank.
5. **BHE'=0, A0=0**: Reads a 16-bit word from an Even address in **1 clock cycle**.
6. Reading an 8-bit byte from Even sets A0=0, BHE'=1 (and vice versa for Odd).
7. Reading a 16-bit word from an **Odd Address** (unaligned) takes **2 clock cycles**.

⚡ Quick Recall

Memory Banking → 2 Banks (Even/Odd) → Even Bank (D0-D7, A0=0) → Odd Bank (D8-D15, BHE'=0)
→ Both=0 = 16-bit transfer in 1 cycle → Unaligned Word = 2 cycles